MTE 204 – Numerical Methods
Tutorial #4 Questions

Question 1:

An investigator has reported the data tabulated below for an experiment to determine the growth rate of bacteria k (per day) as a function of oxygen concentration c (mg/L). It is known that such data can be modeled by the following equation:

$$k = \frac{k_{max} c^2}{c_s + c^2}$$

Where c_s and k_{max} are constants.

$\boldsymbol{c}$	0.5	0.8	1.5	2.5	4
$\boldsymbol{k}$	1.1	2.4	5.3	7.6	8.9

(a) Use a transformation to linearize this equation
(b) Use linear regression to estimate c_s and k_{max}
(c) Predict the growth rate at c = 2 mg/L
(d) Calculate the coefficient of determination (r^2) and the sum of the squares of the residuals (S_r)

Question 2:

The following data points were measured for an engineering experiment:

$\boldsymbol{x}$	1.6	2	2.5	3.2
$\boldsymbol{f(x)}$	2	8	14	15

(a) Develop quadratic splines for the data points and predict $f(2.2)$
(b) Compare your prediction to the prediction of a first-order spline